THE TESTING OF MODELS FOR UNIMOLECULAR DECOMPOSITION VIA INVERSE LAPLACE TRANSFORMATION OF EXPERIMENTAL RECOMBINATION RATE DATA

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The inversion of experimental temperature-dependent recombination rate data, to obtain microcanonical rate constants, is discussed and applied to methyl radical recombination. It is demonstrated that inverse Laplace transformation provides an efficient means of comparing experimental data with model calculations.

1. Introduction

Several models for unimolecular decomposition reactions, occurring on type I potential energy surfaces (i.e. without a potential energy barrier) have been developed, largely by Quack and Troe [1–3], following earlier work by Bunker and Pattengill [4]. Comparison with experiment is achieved by the adjustment of parameters describing the dependence of the potential energy and of certain vibrations on bond distance, and satisfactory agreement can generally be obtained with realistic parameters. Some comparisons have been made which are based on ab initio potential energy surfaces with no adjustable parameters; these latter calculations have sought only a qualitative agreement [5].

Precise experimental techniques are available for measuring the rates of combination reactions and provide a good testing ground for unimolecular rate theory. The rate constants depend only weakly on temperature and so wide temperature ranges can be covered using the same technique, which is not possible for unimolecular dissociation reactions. However, attempts to model the temperature dependence have been very limited and have primarily consisted of demonstrations that certain models, or types of model, reproduce the experimental data more closely than others. To some extent, this reluctance to attempt a more exacting comparison derives from the fact that the more detailed – and successful – models are microcanonical and calculate the energy dependent dissociation rate constant, $k(E)$, whilst experiments measure the canonical rate constant, $k(T)$. Model testing relies on the comparison of the high-pressure limiting experimental rate constant, $k_r^\infty(T)$ with the integral of $k(E)$ over a Boltzmann distribution. Iterative adjustment of the model parameters is clumsy and time consuming and the final optimised results are difficult to interpret. This paper demonstrates the use of the inverse Laplace transformation of recombination data for methyl radicals to provide experimentally based estimates of $k(E)$ which can then be compared more directly with model predictions.

2. Inverse Laplace transformation of recombination rate data

Inverse Laplace transformation of the limiting high-pressure rate constant for *dissociation*, $k_d^\infty(T)$, to obtain $k(E)$, was first proposed by Slater [6] and its application has been discussed in detail by Forst [7–9] and Pritchard [10].

$$k(E_R)N_R(E_R) = \mathcal{L}^{-1}\{Q_R^{vr}(\beta)k_d^\infty(\beta)\}\,, \qquad (1)$$

where E_R is the rovibrational energy of the reactant, R, $N_R(E_R)$ is the rovibrational density of states, Q_R^{vr} the rovibrational partition function and $\beta = 1/kT$.

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It is difficult to make precise measurements of dissociation rate constants over a wide range of temperatures, because of the strong dependence of $k_d^\infty(T)$ on T. Such a limitation is much less pronounced for measurement of the reverse recombination rate constant, $k_r^\infty(T)$, and the development of an expression for recombination rate data, equivalent to eq. (1), is of practical significance.

Consider the reaction

$$A \underset{k_r}{\overset{k_d}{\rightleftharpoons}} B + C ,$$

then

$$k_d^\infty(T) = k_r^\infty(T)[Q_P(T)/Q_R(T)]\exp(-\Delta E_0^0/kT) ,$$

where Q_P and Q_R are the total partition functions for the reactants and products and ΔE_0^0 is the difference in zero point energy between reactants and products.

Defining E_P as the rovibrational energy of the products, relative to their zero point energy, expressing $k_r^\infty(T)$ in Arrhenius form:

$$k_r^\infty(T) = A_r^\infty \exp(-E_r^\infty/kT),$$

and substituting in eq. (1) gives

$$k(E_R)N_R(E_R) = A_r^\infty C' \mathcal{L}^{-1} \int_0^\infty N_P(E_P)(kT)^{3/2} \times \exp[-(E_P + E_r^\infty + \Delta E_0^0)/kT]\, dE_P ,$$

where $C' = (2\pi m_B m_C/h^2 m_A)^{3/2}$ and $N_P(E_P)$ is the total rovibrational density of states for the products B and C. Thus,

$$k(E_R)N_R(E_R) = A_r^\infty C' \int_0^\infty N_P(E_P) \times \mathcal{L}^{-1}\{(kT)^{3/2} \exp[-(E_P + E_r^\infty + \Delta E_0^0)/kT]\}\, dE_P,$$

or, inverting,

$$k(E_R)N_R(E_R) = (2A_r^\infty C'/\pi^{1/2}) \int_0^{E_R - E_r^\infty - \Delta E_0^0} N_P(E_P) \times [E_R - (E_P + E_r^\infty + \Delta E_0^0)]^{1/2}\, dE_P ,$$

$$\text{for } E_R \geqslant E_r^\infty + \Delta E_0^0 . \qquad (2)$$

For our purposes, the main application of the inversion is to provide experimentally based estimates of $k(E_R)$, which can then be compared with rate constants calculated from models. The validity of the inversion was tested by calculating the energy-dependent dissociation constant, $k'(E_R)$, using the minimum sum of states model described in section 3, with realistic parameters for C_2H_6. $k_r^\infty(T)$ was then calculated from $k'(E_R)$, by summing over a Boltzmann distribution, over the temperature range $300 \leqslant T \leqslant 800$ K. Inversion of $k_r^\infty(T)$ produced estimates of the microcanonical rate constants, $k''(E_R)$, which were then compared with the original values $k'(E_R)$. Good agreement was obtained, with fractional differences $\leqslant 10\%$ over the energy range $\Delta E_0^0 + 14000 > E_R > \Delta E_0^0$ cm^{-1}.

3. Application to methyl radical recombination

We recently reported measurements of the rate constant for methyl radical recombination, over the temperature range 298–577 K [11,12]. The limiting high-pressure rate constant decreases slightly with increasing temperature and may, within experimental error, be expressed in the form

$$k_r^\infty = (4.09 \pm 0.36) \times 10^{-11} \exp[(137 \pm 33)\mathrm{K}/T] \ \mathrm{cm^3\ molecule^{-1}\ s^{-1}} . \qquad (3)$$

The densities of states were evaluated over the energy ranges $30300 \leqslant E_R \leqslant 44300$ cm^{-1} and $0 \leqslant E_P \leqslant 14000$ cm^{-1} using an extended Beyer–Swinehart algorithm with a grain size of 10 cm^{-1} and the resulting experimental microcanonical dissociation rate constants, $k(E)$, are shown in fig. 1 (see appendix for spectroscopic data employed; harmonic oscillators were assumed throughout).

Microcanonical rate constants were also evaluated using a variational technique, in which the energy dependent transition state was located by minimising the sum of rovibrational states [13]. Oscillators which change their frequencies significantly on dissociation (defined as soft oscillators by Quack and Troe [1]) were assigned an exponential dependence on bond distance, r (see appendix), whilst the rotational constants were evaluated explicitly as a function of r, for an assumed r dependence of ∠HCC based on the calculations of Duchovic et al. for CH_4 [14]. The transition states are located at quite large bond distances (3.7–5.4 Å) and the molecular torsion was as-

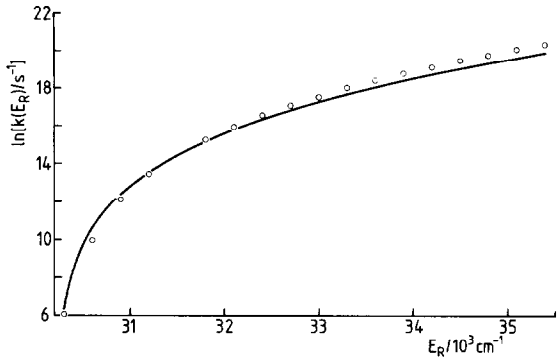

Fig. 1. Microcanonical rate constants as a function of reactant rovibrational energy, E_R. —, $k_{inv}(E_R)$ obtained by inverting the experimental canonical rate constants, $k_r^\infty(T)$. ○, calculated rate constants, $k_{calc}(E_R)$, for a Morse potential function.

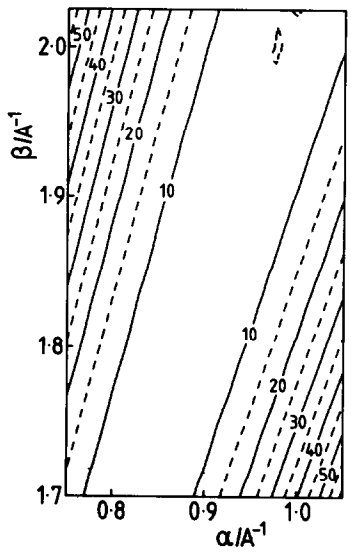

Fig. 2. Σ_1^2 (=Σ [ln $k_{inv}(E_R)$ − ln $k_{calc}(E_R)$]^2) surface for a Morse potential, as a function, of α and β.

sumed to have become a free internal rotor. Both the sums of states and the molecular densities of states were evaluated by direct count and angular momentum was conserved by only including in $N_R(E_R)$ those J states which can dissociate at energy E_R. A comparison of this technique with the more time consuming adiabatic channel model shows good agreement for the range of activated complex C–C distances found here, although the agreement deteriorates for significantly looser complexes [13].

A comparison between inverted, $k_{inv}(E_R)$ and calculated $k_{calc}(E_R)$ microcanonical rate constants was made for a Morse potential energy function, with the Morse parameter, β, and α which describes the distance dependence of the soft frequencies [1] as variable parameters. The best fit was determined by evaluating $\Sigma_1^2 = \Sigma\,[\ln k_{inv}(E_R) - \ln k_{calc}(E_R)]^2$, where the summation was taken over the range $30500 \leqslant E_R \leqslant 32750$ cm^{-1}, with a 225 cm^{-1} grain size, and searching for a minimum in the $(\Sigma_1^2, \alpha, \beta)$ surface. Fig. 2 shows a contour plot for this surface and demonstrates the strong correlation between α and β. The valley is fairly flat and a range of "best-fit" α, β pairs could be obtained; the final choice was made by searching in detail the region around the spectroscopic value of β, giving the best-fit values shown in table 1. The canonical rate constants, $k_r^\infty(T)_{calc}$ are compared with the experimental values, predicted by eq. (3), in table 1.

This method of fitting weights the high-energy rate constants too strongly and an alternative procedure was investigated, in which $\Sigma_2^2 = \Sigma\{W(E_R)[k_{exp}(E_R) - k_{calc}(E_R)]\}^2$ was minimised, where the weighting factors, $W(E_R) = N_R(E_R)\exp(-E_R/k\langle T\rangle)$, with $\langle T\rangle$ = 450 K, the mean of the experimental temperatures, were designed approximately to reflect the contribution of a given energy, E_R, to the canonical rate constant. The resulting best-fit rate constants are shown in fig. 1 and table 1.

The calculated microcanonical rate constants show a significantly stronger energy dependence than the inverted experimental values and $k_r^\infty(T)_{calc}$ rises with temperature, in contrast to the experimental values. An alternative potential energy function was investigated in an attempt to improve the agreement. At the comparatively long bond distances involved, van der Waals forces are likely to be significant and the Morse function inadequate. If it is assumed that the major interaction occurs between the carbon atoms, then the term in r^{-5} will dominate the long-range multipolar interaction. Accordingly, the potential function

$$V(r) = D_e\{1 - \exp[-\beta(r - r_e)]\}^2 - C_5F(r)/r^5$$

was adopted, where $F(r)$ is a damping function:

$$F(r) = 1 - \exp[-(r/r_e - 1)^2] \quad \text{for } r > r_e\,,$$

$$= 0 \quad \text{for } r_e \leqslant r_e\,,$$

Table 1
Experimental and best-fit model values for $k_r^{\infty}(T)$

T (K)	Experimental [a]	Calculated			
		Morse potential		Morse potential + C_5/r^5	
		unweighted $\alpha = 0.89$ Å^{-1} $\beta = 1.815$ Å^{-1}	Boltzmann weighted $\alpha = 0.855$ Å^{-1} $\beta = 1.815$ Å^{-1}	unweighted $\alpha = 0.675$ Å^{-1} $C_5 = 3.05\times10^6$ Å^5 cm^{-1}	Boltzmann weighted $\alpha = 0.6625$ Å^{-1} $C_5 = 3.05\times10^6$ Å^5 cm^{-1}
300	6.46	7.04	4.66	6.41	5.35
400	5.76	8.54	5.80	7.22	6.08
500	5.38	9.61	6.69	7.62	6.48
600	5.14	10.55	7.44	7.96	6.81

[a] Calculated using the experimental Arrhenius expression (eq. (3)).

where r_e is the equilibrium bond distance. C_5 was taken as an adjustable parameter whilst β was fixed at its spectroscopic value. Once again, Σ_1^2 and Σ_2^2 were minimised by searching the (Σ^2, α, C_5) surface. A strong correlation was found once more, and the search was concentrated around the $C_5 \approx 3 \times 10^6$ Å^5 cm^{-1} region, which is the value estimated by Chang for two ^{2}P carbon atoms [15]. The ratio of the best-fit calculated microcanonical rate constants to the inverted experimental values, $k_{calc}(E_R)/k_{inv}(E_R)$ is shown in fig. 3 as a function of energy, whilst the canonical rate constants are included in table 1. The fit is improved, but there is still a substantial difference between the energy dependences of the calculated and inverted rate constants. Fig. 3 includes error bars, which reflect the uncertainties in the experimental activation energy, showing that the energy dependence required by the model lies well outside experimental uncertainty. Even removing the temperature dependence of the experimental rate constant failed to provide good agreement, although obviously the fit was much improved.

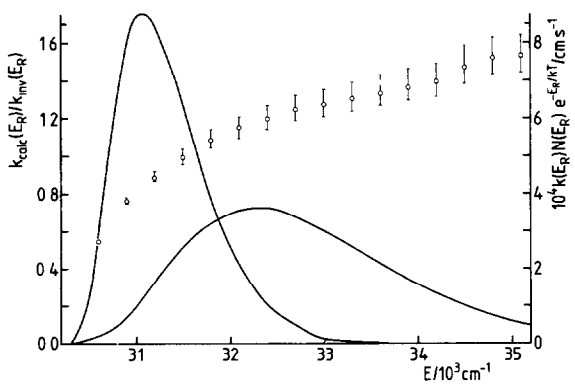

Fig. 3. A plot of $k_{calc}(E_R)/k_{inv}(E_R)$ versus E_R for a Morse + C_5/r^5 potential function. The error bars represent the uncertainty in $k_{inv}(E_R)$ arising from the uncertainty in E_r^{∞} (2σ). The curves represent the Boltzmann functions, $N(E_R) \times \exp(-E_R/kT)$ at the extremes of the experimental temperature range, 300 and 600 K.

4. Discussion

The main aim of this work has been to investigate the inversion of recombination rate data to obtain microcanonical rate constants which may then be compared with model calculations. The success, or otherwise, of any particular model, or parameter set, is, therefore, of secondary consideration, although we make some brief comments on this aspect below.

The data shown in table 1 demonstrate that a comparison at the microcanonical level produces values for $k_r^{\infty}(T)_{calc}$ which, when averaged over temperature, closely match the experimental values. This agreement is significantly improved if the Boltzmann weighting procedure is employed. The failure of the calculations to match the experimental temperature dependence represents a failure of the models used, rather than of the inversion procedure.

The calculations also demonstrate the operational ease of fitting model parameters to experimental recombination data via an inverse Laplace transforma-

tion. A single transformation produces microcanonical rate constants, which may then be compared with the model calculations. A comparison via the canonical rate constants themselves is less easy; the $k(E)$ values calculated for each parameter set must be integrated over a Boltzmann distribution, at each temperature, to obtain a set of $k_r^\infty(T)_{calc}$. Searching for the best-fit values, is, therefore, more time consuming. Furthermore, the reasons for the failure of a given model are more transparent if the comparison is made at the microcanonical level. The Arrhenius form for $k_r^\infty(T)$, assumed in the present treatment, need not necessarily be employed – more complex functional forms may be transformed analytically, or numerically if required. The inversion is only performed once and is not a component of a repetitive fitting procedure.

Fig. 3 shows the form of the Boltzmann distribution function, $N_R(E_R)\exp(-E_R/kT)$ for $T = 300$ and 600 K and demonstrates the range of energies for which significant experimental information on $k(E)$ is obtained. Attempts to match experiment and theory outside, or at the wings, of this energy range are clearly invalid. In particular the data, at these temperatures, give little information on the form of $k(E)$ close to the threshold, as is clearly demonstrated by eq. (1), which predicts that $k(E)$ is non-zero significantly below the threshold energy, because E_r^∞ is negative. Clearly the Arrhenius expression must become invalid at lower temperatures.

Although inversion permits a comparison between theory and experiment at the microcanonical level, it is subject to two general limitations. Firstly, as implied above, the inversion is only unique if an infinite range of temperature is covered. An extension of the experimental range would be valuable, especially to lower temperatures, where $k(E)$ close to threshold becomes more significant. Attempts to extend the high temperature range by employing dissociation data are of limited utility because of the comparatively large uncertainty in the enthalpy of dissociation which, even for $C_2H_6/2CH_3$ is at least comparable with the negative activation energy for recombination.

The second limitation is related to the rotational averaging implicit in the experimental data. Microcanonical calculations, such as those based on the adiabatic channel model [1], or the variational treatment of Wardlaw and Marcus [16], show a strong dependence of the rate constant on the rotational quantum model J, especially near threshold. A particularly strong J dependence has been demonstrated in the recent calculations by Clary [17] on rate constants for ion-dipole reactions. This level of detail is not easily probed by thermal rate constants, even at low temperatures, for molecules with small B values, such as C_2H_6. Low-temperature data on, for example, CH_3 + H, might prove more discriminating.

Finally, we make two points on the fitting to the methyl radical recombination data.

(i) The models clearly and consistently overestimate the energy dependence of $k(E)$ and it is of interest to examine the experimental evidence for the negative temperature dependence of $k_r^\infty(T)$. These experimental, limiting high-pressure rate constants were obtained by extrapolation using the Troe factorisation technique [18]; the ranges of the extrapolations are quite small, but increase with increasing temperature (7% of k_r^∞ at 296 K to 16% at 577 K). Some more detailed calculations were also performed, using the microcanonical model referred to above [13] and an energy grained master equation to model the fall-off behaviour, but these calculations also produced a negative value for E_r^∞. Further measurements, especially at low temperatures, would be of value.

(ii) If we accept that E_r^∞ is indeed negative, then the model employed above seriously overestimates the energy dependence of $k(E)$. Accepting the basic premises of RRKM theory, there are three possible reasons for this failure. Firstly, the long-range attraction, which will enhance $k(E)$ at low E, but have little effect at high E, may be inadequately represented. The inclusion of the r^{-5} term is, perhaps, insufficient and higher inverse powers might contribute significantly. Secondly, the angular dependence of the potential may play an important role. As the energy increases, higher vibrational levels in the CH_3 rocking modes become populated and, since these modes correlate with CH_3 rotations, a high-amplitude rocking motion will occur in the complex, which will interact strongly with, and perhaps even hinder, the internal rotation about the C–C axis. Finally, the calculations are based entirely on harmonic sums and densities of states. Both $k_{inv}(E_R)$ and $k_{calc}(E_R)$ contain $N_R(E_R)$, the density of states of the reactant molecule at energy E_R, as a denominator (see eq. (1) and, for example, ref. [19]) and errors in this quantity do not, therefore, affect the present comparison.

The numerator for $k_{calc}(E_R)$ contains the sum of states for the complex [19], whilst that for $k_{inv}(E_R)$ contains, in effect, the sum of states for the separated radicals, weighted by a slowly varying energy function (eq. (1)). Neither of these sums refers to large amounts of energy and the quantum numbers involved are small, thus reducing the effects of anharmonicity. Corrections could be readily made to $k_{inv}(E_R)$, thus increasing it, especially at higher energies. It is more difficult to make similar corrections to $k_{calc}(E_R)$, because the shape of the potential energy surface in the region of the complex is unknown. One might expect that anharmonicity would be greater in the complex than in the separated reactants; its inclusion would then lead to a more rapid increase with energy for $k_{calc}(E_R)$ than for $k_{inv}(E_R)$, thus exacerbating the disagreement.

We conclude that the model we have employed, whilst giving qualitative agreement with experiment, is not capable of a detailed description of the reaction. An improved temperature dependence is found with the canonical rate constants calculated from the adiabatic channel model [20], but the significant decrease with temperature is not reproduced. The difficulty of modelling the reaction is of practical significance since it means that rate constants calculated at, say, flame temperatures, based on our experimental data and the best fit model, will be of limited accuracy. Detailed ab initio calculations of the angular and radial potential, to high C–C bond distances, would be of value. Application of the unimolecular reaction model recently developed by Wardlaw and Marcus [16], which can handle angular potential functions and anharmonicity, using realistic potential functions, would be of great interest.

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Appendix. Molecular parameters for $C_2H_6/2CH_3$

Vibrational frequencies (cm^{-1}):

C_2H_6	CH_3
2985(2)	3162(2)
2969(2)	
2954	3044
2896	
1469(2)	1396(2)
1468(2)	
1388	606
1379	
1190(2)	
995	reaction coordinate
822(2)	
298	rotation

Exponential variation of soft vibrational frequencies with C–C bond length, r:

$\nu^+ = \nu_R \exp[-\alpha(r - r_{eq})] + \nu_P \{1 - \exp[-\alpha(r - r_{eq})]\}$,

ν^+ = vibrational frequency of complex,

ν_R, ν_P = vibrational frequencies of reactant and product, respectively.

Rotational constants (cm^{-1}):

$C_2H_6: A = 2.671, B = C = 0.6625.$

$CH_3: A = 4.789, B = C = 9.578.$

Variation of rotational constants (cm^{-1}) with C–C bond length, r (Å):

$A = 3.813 - 1.120r + 3.187 \times 10^{-1} r^2 - 3.971 \times 10^{-2} r^3 + 1.835 \times 10^{-3} r^4.$

$B = 2.143 - 1.533r + 4.530 \times 10^{-1} r^2 - 6.119 \times 10^{-2} r^3 + 3.107 \times 10^{-3} r^4.$

Morse parameter, $\beta = 1.815$ Å^{-1}.

$\Delta E_0^0 = 30320$ cm^{-1}.

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